The Impact of Household Surveys on 2020 Census Self-Response

by

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Abstract

Households who were sampled in 2019 for the American Community Survey (ACS) had lower self-response rates to the 2020 Census. The magnitude varied from -1.5 percentage point for household sampled in January 2019 to -15.1 percent point for households sampled in December 2019. Similar effects are found for the Current Population Survey (CPS) as well.*

Keyword*:* nonresponse, response rates

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* Author contact: jonathan.s.eggleston@census.gov. This report is released to inform interested parties of ongoing research and to encourage discussion of work in progress. Any opinions and conclusions expressed herein are those of the author and do not reflect the views of the U.S. Census Bureau. The Census Bureau's Disclosure Review Board and Disclosure Avoidance Officers have reviewed this data product for unauthorized disclosure of confidential information and have approved the disclosure avoidance practices applied to this release. DRB Approval Numbers: #CBDRB-FY22-116 and CBDRB-FY22-271.

Introduction: To minimize the cost of the 2020 Census, much effort was spent on maximizing self-response rates for 2020, particularly encouraging people to respond online. Unfortunately, the ability to move respondents to internet response is hampered by limited knowledge in the survey field on what factors effect someone's decision to participate in surveys. For example, while it's well known that survey response rates have been declining over time, it's relatively unknown the reasons for this decline (Czajka and Beyler 2016). One proposed explanation for the declining response rates is the recent increase in the number of surveys, both inside the federal statistical systems and in the private sector over the past few decades (Leeper 2019).

In the context of the decennial census, this could mean that households who were recently sampled for a survey might feel an extra sense of burden when they are contacted to complete the 2020 Census. Further, given this sense of burden, these households may be less likely to self-respond to the 2020 Census. Comparing the American Community Survey (ACS) self-response around the 2010 Census, Baumgardner (2013) finds that ACS self-response rates increased January through March, but then decreased April through June. While this is suggestive that ACS or other surveys may affect decennial self-response rates, to date, no research has been done to evaluate this.

This paper explores this hypothesis by linking the listing of addressed sampled for the ACS and Current Population Survey (CPS) between 2015 and 2019 to the 2020 Census Unedited File. From this linking, we find that households who were sampled in 2019 for the ACS have lower self-response rates to the 2020 Census. The magnitude varies from -1.5 percentage point for household sampled in January 2019 to -15.1 percent point for households sampled in December 2019. Similar effects are found for the CPS. While respondent fatigue may be a likely explanation for this effect, there could be other factors at play, such as confusion about the differences between the decennial census and household survey sponsored by the U.S Census Bureau. Nevertheless, these results are highly suggestive that prior survey participation affects the decision to self-respond to the decennial census, which has implications for refusal conversation and communication campaigns in future decennial censuses.

Methodological Detail

The decennial census, American Community Survey, and the Current Population Survey are valuable sources for determining basic facts about the U.S. Population. The decennial census provides a complete count of the total population for states and local areas, while the ACS provide information on the demographic and economics characteristics of the nation as well as local areas, such as the percent of residents in a county living in poverty. The CPS is the primary source of labor force statistics, such as the unemployment rate, for the United States.

The main start date of the ACS was January 2005, when all areas of country became eligible for the survey.[1] The ACS releases estimates annually, but interviews households continuously throughout the year. The ACS consists of twelve independent monthly samples. The month an address can first respond to the ACS is referred to as the ACS *panel;* each address or household is given up to three months to respond to the survey. The Census Bureau's Master Address File (MAF) is the sampling frame

[1] The ACS was tested prior to 2005 in some counties across the US. Group Quarters (e.g., nursing homes and college dorms) were added to the survey in January 2006.

for the ACS, CPS, and other household surveys. The MAF is also used as the enumeration universe for the decennial census. Given this, ACS, CPS and 2020 Census addresses can be linked together at the address level by the key identifier in the MAF, called the MAFID.

The outcome of interest in this analysis is whether a household self-responded to the 2020 Census.[2] The key explanatory variables are indicators for year and month a household was sampled for the ACS or CPS. Note that not all areas of the country receive the same eligibility to self-respond. In some rural areas of Alaska and Maine, households are enumerated in person from the start. Because of this factor and an additional detail on address quality discussed in the next paragraph, a sample restriction of only including households in the "self-response" type of enumeration area (TEA value of 1) is employed.

In addition, I only include household units that were occupied for the 2020 Census, as well as only addresses that were in the Mater Address File at the start of 2020 Census operations. In the Census, address can be added after field operations starts, but I exclude these addresses since they could not have been sampled for a household survey earlier in the decade.

There are some additional complications which could confound the results. First, for the ACS, some areas of the county with low internet connectivity receive a mail questionnaire first rather than an initial invitation to respond online, limiting the possible ways they can self-respond. Second, not all areas in the United States are sampled for the ACS with equal probability. In particular, the ACS oversamples counties with small populations and areas with a higher predated self-response rate. Thus, the odds of being sampled for ACS are likely related to one's propensity to self-respond the 2020 Census. In the CPS, most addresses in a state have the same overall probability of selection (U.S. Census Bureau 2019), so this issue is less of a problem for the CPS. Finally, the U.S. Census Bureau might not receive a response from an address if there is an error in the household's listed address, resulting in the post-office being unable to deliver the household survey or 2020 Census invitation. During ACS nonresponse follow-up and CPS operations, any errors found in the household's address listed on the MAF are corrected. Thus, households sampled in ACS and CPS might have slightly higher odds of successfully receiving a decennial census invitation than the rest of the county.

To address these issues, our primary specification is a fixed effects regression with block fixed effects. In this regression, the dependent variable is whether a household self-responds to the 2020 Census, and the explanatory variables are only the block fixed effects and a series of indicators for whether a household was sampled for a household survey in a given month and year. Formally, with Y_{ib} denoting whether household i in block b self-responds to the 2020 Census, and S_{ibtm} denoting whether the household was sampled for the ACS or CPS in year t and month m, the fixed-effects model is specified as

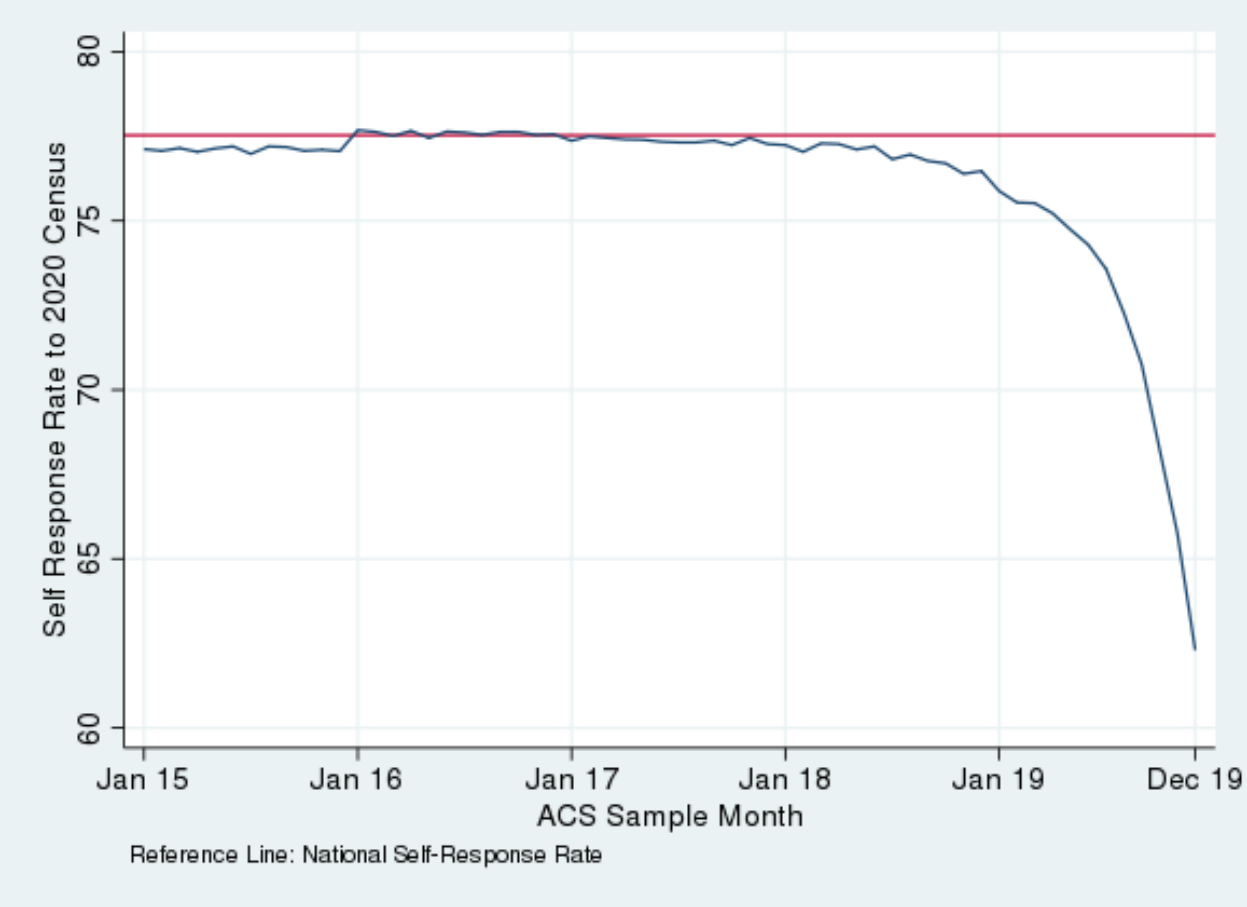
$$Y_{ib} = \sum_{t=2015}^{2019} \sum_{m=1}^{12} \beta_{tm} S_{ibtm} + \alpha_b + u_{ib}$$

[2] Self-response is defined as households who responded online, mailed in a paper form, or took the initiative to respond via phone through the Census Questionnaire Assistance Program.

Because ACS oversampling determination are done at the block level, this should account for factors related to the ACS sampling design. In the CPS, blocks are also the smallest level of geography used in the sampling procedures. Thus, this specification controls for sampling rate differences in the CPS as well. And given address quality is highly related to the rurality of an area, having block fixed effects and only including households in the "self-response" designated areas should mostly mitigate issues with differential address data quality. Thus, the fixed-effects regression should represent the effect of being sampled in ACS or CPS on the probability of internet self-response for the 2020 Census.

Figure 1: 2020 Census Self-Response Rates for Occupied Housing Units:

by Year and Month Sampled for ACS



Source: 2020 Decennial Census Unedited File and 2015 to 2019 American Community Survey Address File. Figure shows self-response rates to the 2020 Census for addresses recently sampled for the ACS, by month sampled. National Self-Response in this chart may not match other published census numbers given sample restrictions.

Figure 2: Fixed Effects Coefficients Estimates Rates by Year and Month Sampled for ACS



Source: 2020 Decennial Census Unedited File and 2015 to 2019 American Community Survey Address File. Figure plots coefficients from a regression model where the dependent variable is whether a household self-responds to the 2020 Census, and the explanatory variables are only the block fixed effects and a series of indicators for whether a household was sampled for the ACS in a given month and year. Figure shows the estimates of the ACS indicators.

Results

Figure 1 shows the response rates by month and year a household was sampled for ACS, for years 2015 to 2019. Figure 2 shows the coefficient from our fixed effect regression, where each point represents the difference in response odds between household sampled in ACS in the given month from the base outcome of household who were not in ACS anytime between 2015 and 2019. Overall, the two figures show similar trends of response rates decreasing rapidly throughout 2019. The magnitude varies from -1.5 percentage point for household sampled in January 2019 to -15.1 percent points for households sampled in December 2019. Going further back in time, there is a slight negative effect in late 2018 that is less than one percentage point. For the most parts, the effects are insignificant for 2017 and 2016, with a slight negative effect in 2015.

Figure 3: Fixed Effects Coefficients Estimates Rates by Year and Month Sampled for CPS



Source: 2020 Decennial Census Unedited File and 2015 to 2019 Current Population Survey. Figure plots coefficients from a regression model where the dependent variable is whether a household self-responds to the 2020 Census, and the explanatory variables are only the block fixed effects and a series of indicators for whether a household was sampled for the CPS in a given month and year. Figure shows the estimates of the CPS indicators.

Current Population Survey

The previous figures show a negative effect of being recently sampled for the ACS on 2020 self-response. One concern could be that this effect from the ACS is unique and not generalizable to other surveys a household could receive. The ACS and the 2020 Census have multiple similarities, given both the ACS and 2020 Census are mandatory surveys and both have an initial push for self-response before in-person follow-up. To address this concern, I also estimate this model on the CPS. CPS interviews are also conducted by the Census Bureau, so there are still contact strategy similarities between the CPS and the 2020 Census, such as receiving notification letters from the U.S. Census Bureau with similar letterhead. However, the CPS does not have a self-response mode; initial interviews are primarily in person and subsequent interviews are often over the phone. In addition, the CPS is a voluntary survey. Thus, the different characteristics of the CPS compared to the ACS should provide a robustness check to see if the ACS results extends to other household surveys.

Figure 3 shows the results for the fixed-effects model run on the CPS. The results show a similar story, but with some slight complications given the CPS sample design. In the CPS, households are interviewed once a month for four months, followed by an eight-month break, followed by another four-month interview period. Therefore, households sampled in February 2019 were also interviewed in the CPS from February 2020 to May 2020, which is the same time as the initial push for self-response for the 2020 Census. Households sampled in June 2019, on the other hand, wouldn't have been interviewed again in 2020 until June, well after the initial 2020 self-response period. Thus, this may explain why the self-response rate is lower for CPS households first sampled in early 2019 compared to those sampled later in 2019.

Compared to Figure 2 for the ACS, the CPS results show similar point estimates for many months in 2019, with a negative effect of a few percentage points. The estimates are slightly nosier given the CPS's smaller sample size. The CPS has a sample of about 60,000 households per month compared to ACS sampling almost 3 million addresses annually.[3] The largest negative estimates for the CPS are about -8 to -9 percentage points, which is lower than the -15.1 estimate for the December 2019 ACS panel. Therefore, the effects on 2020 self-response from the CPS may not be as strong. Nevertheless, the findings that being sampled for the CPS also shows a negative effect on self-response to the 2020 Census suggests that this may be a universal effect from being sampled in any household survey, and not just a unique feature of the ACS.

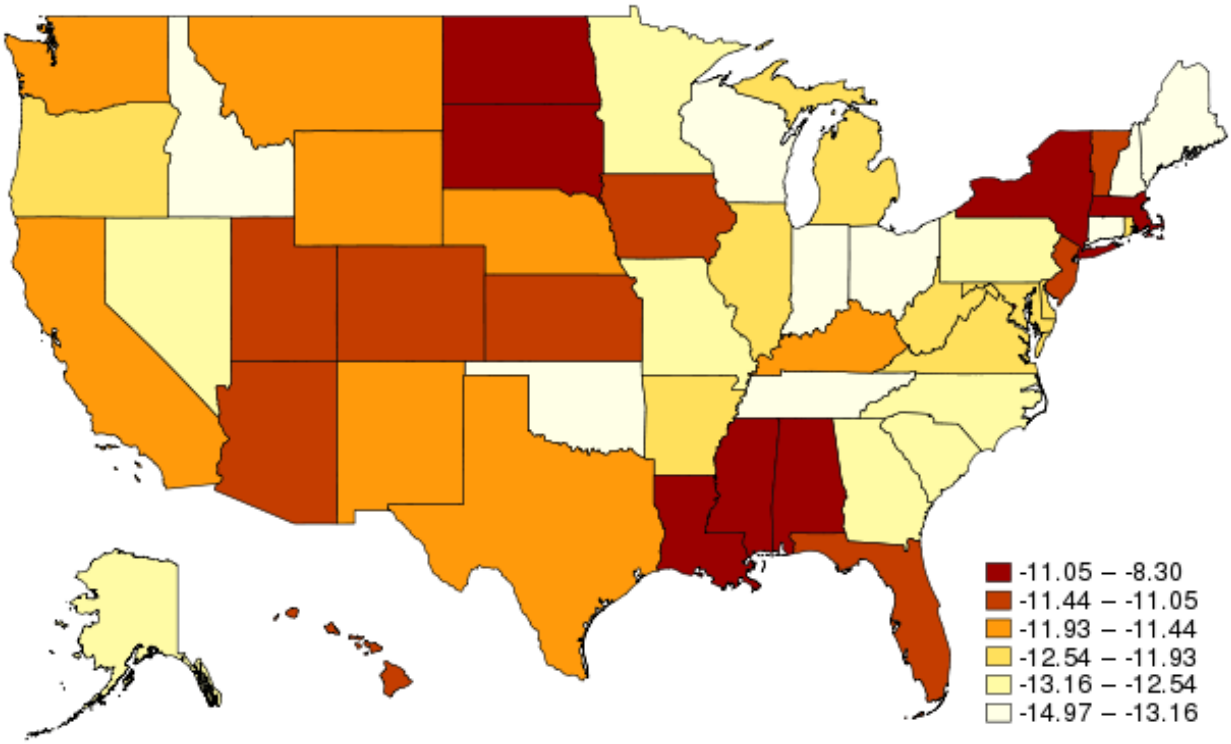
[3] See Appendix Tables A.1 and A.2 for the standard errors of the ACS and CPS estimates.

Table 1-2020 Census Self-Response Rate and ACS Fixed-Effect Estimate by Demographic Groups

Category	Self-Response Rate	Fixed-Effect Estimate:	Fixed Effect S.E.
Any Children	79.73	-10.99	0.10
Any Over 65	89.06	-11.19	0.10
One Adult-No Children	79.22	-10.62	0.10
Two Adults-No Children	86.83	-11.71	0.08
Three or More Adults-No Children	83.66	-10.93	0.17
One Adult-With Children	71.33	-7.97	0.36
Two Adults-With Children	81.73	-11.88	0.13
Three or More Adults-With Children	81.06	-10.19	0.22
Any White Non-Hispanic	87.78	-8.99	0.05
Any Black	75.69	-7.45	0.16
Any Hispanic	75.60	-8.32	0.16
Any Asian	86.30	-5.43	0.21
Any Native Hawaiian Pacific Islander	72.42	-10.43	1.89
Any American Indian Alaska Native	80.20	-9.08	0.69

Source: 2020 Decennial Census Unedited File and 2015 to 2019 American Community Survey. Table presents self-response rates and fixed-effects estimates by demographic groups. The Census Unedited File only contains non-imputed age and race data, and does not contain coding of race write-in data. Fixed-Effects estimate are the estimates from a regression model where the dependent variable is whether a household self-responds to the 2020 Census, and the explanatory variables are only the block fixed effects and a series of indicators for whether a household was sampled for ACS in a given month and year. The fixed-effects estimate shown is the indicator of whether a household was sampled in the October, November, or December 2019 ACS. Sample only included housing units which were occupied for the 2020 Census and were in the Master Address File at the start of 2020 Operations

Figure 4: ACS Fixed-Effects Estimate by State



Source: 2020 Decennial Census Unedited File and 2015 to 2019 American Community Survey. Figure presents self-response rates and fixed-effects estimates by demographic groups. The fixed-Effects estimate are the estimates from a regression model where the dependent variable is whether a household self-responds to the 2020 Census, and the explanatory variables are only the block fixed effects and a series of indicators for whether a household was sampled for ACS in a given month and year. The fixed-effects estimate shown is the indicator of whether a household was sampled in the October, November, or December 2019 ACS.

Variation Across Geography and Demographics

To show how the effect of being sampled in a household survey varies by geography and demographic groups, Table 1 and Figure 4 present some additional analyses. In this section, I run the fixed-effects regression model separately by state and certain demographic groups. I run the model on just the ACS rather than the CPS given the much larger sample size for the ACS. For age, race and ethnicity, I use the standardized versions of the variables available on the Census Unedited File, which excludes and imputed race data or coding of write-in responses. In order to improve the precision of some smaller states and groups, I modify the regression model to combine the indicators of being sampled in the October, November, and December 2019 ACS panels into one indicator, and I present the estimate of that indicator in Table 1 and Figure 4. For table 1, the fixed-effects estimates are relatively stable across groups. Household with one adult and no children as well as household with a member who is Black have lower fixed-effect estimates, but lower self-response rates overall, so perhaps the effects of being sampled for ACS are less impactful for groups with lower response propensities generally. One group that stands out is households with any Asian member, who have a higher self-response rate of 86.3 percent but a fixed-effects estimate of only -5.43.

For geography, there is some variation by state, with fixed effects estimates ranging from -8.26 to 14.99. There is some geographic cluster, such as Louisiana, Mississippi, and Alabama which all have relatively small estimates in term of absolute value. These states also have lower self-response rates, as shown in Table A.1. But overall, there isn't a clear geographic trend. For example, take the Northeast. New York has an estimate of -10.52 and Connecticut has a value of -8.26, while New Hampshire has an estimate of -14.99 and Maine has an estimate of -13.45.

Discussion

The results above present large effects of recent completion of a household survey on decennial self-response. An explanation for these results could be respondent fatigue, with many respondents feeling burdened by having to complete both a household survey and the decennial census in a short timeframe. However, there could be other factors at play. For the ACS in particular, some individuals may not be that aware of the difference between ACS and the decennial census and may have thought they have already responded to the 2020 Census when they completed the ACS. Like the decennial Census, the ACS is sponsored by the Census Bureau, is mandatory, and encourages most households to respond initially via internet.

Stein et al. (2021) conducted interviews of 2020 ACS respondents to see how they remember the mail material. The ACS mail material contained additional language emphasizing the difference between ACS and the 2020 Census. Nevertheless, only about half of respondents remembered the mail materials differentiating between the ACS and the 2020 Census, while the other half didn't remember there being a difference. These latter respondents said they did not pay close attention to the letters and thought that all the material were about the 2020 Census. Similarly, Spiers and Heimel (2021) found that households in the March 2020 ACS sample who received specialized letters distinguishing the ACS from the 2020 Census had a higher self-response rate to both the ACS and the 2020 Census.

To ensure the results were not due to idiosyncratic features of the ACS, I used the CPS as well. The CPS is reasonably difference from the 2020 Census given interviews are in person and over the phone, with no option for a mail or internet response, which should reduce respondent confusion. However, both

surveys are still conducted by the Census Bureau, so the mail material may have still seemed similar between the CPS and 2020 Census, which could lead to respondents thinking they were getting another notification about the CPS. Overall, the results are similar with the CPS, although some of the point estimates are smaller (in absolute value terms) than what is found with the ACS. Given this, the larger effects with ACS may suggest that respondent confusion partially contributes to the ACS results. But finding similar negative affects among two different household surveys suggests additional factors are at play, which may suggest that respondent fatigue is contribution to the lower self-response rates for recent household survey respondents.

In summary, these results are highly suggestive that prior survey participation affects the decision to self-respond to the decennial census. For 2020, these results may be suggestive that special attention may be needed in the Nonresponse Follow-up (NRFU) data collection operation for recent respondents to household surveys, even though these households are a small percentage of the population. Potentially, individuals expressing a feeling of being too busy to respond to the decennial census may be in fact reflecting a belief that they have already provided the Census Bureau the requisite information in a prior survey rather than the respondents not having the time to complete the decennial census questionnaire. For 2030 and beyond, these results suggest that it could be worthwhile to monitor growth trends in the number of federal and non-federal surveys, in order to gauge the prevalence of someone completing a survey right before the decennial census.

In addition, if there is interest in reducing the respondent burden of recent household survey respondents in the context of the decennial census, there could be a way of using the prior survey response information during enumeration. For example, with recent ACS respondents, it is possible to build a model to predict when a household roster reported in the 2009 and 2019 ACS is the same as the roster reported on the 2010 and 2020 Census, respectively, for this same MAFID. From the model, one could use these predicted values to create a cutoff rule of when the ACS roster is likely to be the same as the decennial roster. ACS households that meet this cutoff could receive fewer visits during NRFU. This is similar to the administrative data models used in the 2020 Census. For the 2020 Census, the outcome of interests was the agreement between a decennial roster and a household roster constructed from administrative data, while in this instance the comparison is with an ACS roster. This would require additional research, particularly since the ACS and the decennial census have different residency rules.

While such an analysis is possible, there is a potential complication related to processing schedules. The ACS edited data typically is not finalized typically until August of the following year. For example, the 2029 ACS internal edited data would not be finalized until August of 2030, assuming the current production schedule. Since NRFU typically begins month earlier, these finalized data would not be available during decennial production. However, the unedited data for all cases should be available by March of the following year.[4] Thus, if a special transfer was arranged, it could be possible to use these unedited data for decennial processing. It should be noted that since the number of affected households would make up a small percentage of the U.S Population, the reduction in decennial NRFU workloads would be small. Thus, the main benefit of this procedure would be to reduce potential respondent burden for households selected for the American Community Survey. Such an approach could be extended to the CPS and other surveys conducted by the Census Bureau. But given the

[4] For households sampled in December 2019, for example, nonrespondent households were be interviewed in-person during February 2020, so the unedited data for these household should be available by March.

smallest sample size of these other surveys, the tradeoff between increased operational costs compared to the reduction in respondent burden may be different.

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Appendix

Table A.1 2020 Census Self-Response Rates and ACS Fixed Effects Estimates for Occupied Housing Units, by Month Sampled

Year Sampled	Month Sampled	Self-Response Rate	Fixed-Effect Estimate	Fixed Effect S.E.
2015	1	77.12	-0.37	0.09
2015	2	77.07	-0.43	0.09
2015	3	77.15	-0.28	0.09
2015	4	77.04	-0.42	0.09
2015	5	77.14	-0.34	0.09
2015	6	77.20	-0.27	0.09
2015	7	76.98	-0.50	0.09
2015	8	77.20	-0.28	0.09
2015	9	77.18	-0.27	0.09
2015	10	77.07	-0.39	0.09
2015	11	77.10	-0.35	0.09
2015	12	77.06	-0.42	0.09
2016	1	77.68	0.19	0.09
2016	2	77.63	0.15	0.09
2016	3	77.51	0.03	0.09
2016	4	77.66	0.19	0.09
2016	5	77.45	0.00	0.09
2016	6	77.64	0.16	0.09
2016	7	77.61	0.17	0.09
2016	8	77.54	0.06	0.09
2016	9	77.63	0.16	0.09
2016	10	77.63	0.14	0.09
2016	11	77.54	0.05	0.09
2016	12	77.56	0.10	0.09
2017	1	77.37	-0.08	0.09
2017	2	77.50	0.05	0.09
2017	3	77.45	-0.03	0.09
2017	4	77.41	-0.04	0.09
2017	5	77.40	-0.05	0.09
2017	6	77.34	-0.12	0.09
2017	7	77.32	-0.13	0.09
2017	8	77.32	-0.13	0.09
2017	9	77.37	-0.09	0.09
2017	10	77.24	-0.23	0.09
2017	11	77.45	-0.01	0.09
2017	12	77.27	-0.19	0.09
2018	1	77.24	-0.19	0.09
2018	2	77.04	-0.43	0.09
2018	3	77.29	-0.18	0.09

2018	4	77.27	-0.17	0.09
2018	5	77.11	-0.31	0.09
2018	6	77.20	-0.29	0.09
2018	7	76.82	-0.58	0.09
2018	8	76.96	-0.48	0.09
2018	9	76.77	-0.63	0.09
2018	10	76.70	-0.77	0.09
2018	11	76.39	-1.03	0.09
2018	12	76.47	-0.99	0.09
2019	1	75.88	-1.52	0.09
2019	2	75.54	-1.87	0.09
2019	3	75.52	-1.89	0.09
2019	4	75.22	-2.18	0.09
2019	5	74.74	-2.69	0.09
2019	6	74.29	-3.18	0.09
2019	7	73.57	-3.85	0.09
2019	8	72.27	-5.15	0.09
2019	9	70.75	-6.66	0.09
2019	10	68.24	-9.20	0.09
2019	11	65.76	-11.66	0.09
2019	12	62.30	-15.12	0.09

Source: 2020 Decennial Census Unedited File and 2015 to 2019 American Community Survey Address File. See note for Figure 2 for more details.

Table A.2 2020 Census Self-Response Rates and CPS Fixed Effects Estimates for Occupied Housing Units, by Month Sampled

Year Sampled	Month Sampled	Self-Response Rate	Fixed-Effect Estimate	Fixed Effect S.E.
2015	1	77.65	0.18	0.49
2015	2	77.23	-0.30	0.49
2015	3	77.32	-0.39	0.49
2015	4	77.47	-0.10	0.50
2015	5	77.34	0.00	0.50
2015	6	77.11	-0.69	0.50
2015	7	76.62	-0.79	0.50
2015	8	76.60	-1.29	0.50
2015	9	78.26	0.31	0.50
2015	10	78.64	0.84	0.50
2015	11	77.64	-0.53	0.50
2015	12	78.18	0.27	0.49
2016	1	77.01	-0.78	0.50
2016	2	77.65	-0.08	0.49
2016	3	77.42	-0.67	0.50
2016	4	77.42	-0.22	0.49
2016	5	76.62	-0.90	0.49
2016	6	76.58	-1.60	0.49
2016	7	77.84	0.03	0.49
2016	8	76.39	-0.84	0.49
2016	9	76.40	-1.41	0.49
2016	10	76.56	-1.34	0.49
2016	11	77.68	-0.54	0.49
2016	12	76.04	-1.85	0.49
2017	1	75.91	-1.93	0.49
2017	2	76.20	-1.57	0.49
2017	3	76.14	-1.56	0.49
2017	4	76.85	-0.64	0.50
2017	5	76.32	-1.27	0.50
2017	6	76.61	-1.37	0.50
2017	7	75.73	-1.84	0.50
2017	8	76.19	-1.61	0.50
2017	9	75.48	-2.11	0.50
2017	10	76.35	-1.62	0.50
2017	11	76.07	-1.80	0.50
2017	12	75.76	-1.74	0.50
2018	1	75.15	-2.67	0.50
2018	2	75.86	-1.72	0.50
2018	3	73.99	-3.03	0.50
2018	4	75.68	-2.11	0.50

2018	5	74.13	-3.74	0.50
2018	6	74.13	-3.49	0.50
2018	7	73.65	-3.87	0.50
2018	8	73.41	-4.61	0.50
2018	9	73.00	-4.68	0.50
2018	10	73.39	-4.36	0.50
2018	11	72.06	-5.38	0.50
2018	12	69.63	-7.78	0.50
2019	1	69.26	-7.90	0.50
2019	2	69.17	-8.38	0.50
2019	3	69.33	-8.16	0.50
2019	4	73.92	-3.73	0.51
2019	5	75.51	-2.55	0.51
2019	6	74.30	-3.22	0.51
2019	7	74.84	-3.10	0.50
2019	8	73.49	-4.38	0.51
2019	9	73.39	-4.37	0.51
2019	10	72.21	-5.49	0.51
2019	11	70.28	-7.35	0.51
2019	12	68.78	-9.34	0.51

Source: 2020 Decennial Census Unedited File and 2015 to 2019 Current Population Survey. See note for Figure 3 for more details.

Table A.3 2020 Census Self-Response Rates and ACS Fixed Effects Estimates for Occupied Housing Units, by State

Category	Self-Response Rate	ACS Fixed-Effect Estimate	ACS Fixed Effect S.E.
Alabama	78.27	-9.96	0.39
Alaska	72.92	-12.95	1.42
Arizona	76.70	-11.23	0.36
Arkansas	76.97	-12.20	0.53
California	76.86	-11.69	0.16
Colorado	80.25	-11.26	0.37
Connecticut	77.60	-13.17	0.48
Delaware	76.71	-12.28	0.89
District of Columbia	75.39	-9.04	1.06
Florida	75.90	-11.32	0.21
Georgia	73.89	-12.58	0.32
Hawaii	76.77	-11.39	0.83
Idaho	80.73	-13.16	0.72
Illinois	79.97	-12.01	0.23
Indiana	79.65	-13.55	0.33
Iowa	81.20	-11.12	0.40
Kansas	80.13	-11.35	0.46
Kentucky	80.11	-11.91	0.40
Louisiana	73.80	-10.99	0.45
Maine	76.36	-13.57	0.69
Maryland	78.62	-11.97	0.36
Massachusetts	78.07	-8.30	0.36
Michigan	83.21	-12.10	0.22
Minnesota	84.49	-12.84	0.27
Mississippi	74.58	-10.57	0.59
Missouri	80.43	-12.63	0.33
Montana	79.82	-11.57	0.84
Nebraska	81.15	-11.62	0.52
Nevada	74.45	-12.98	0.54
New Hampshire	79.00	-14.97	0.75
New Jersey	77.52	-11.38	0.31
New Mexico	76.23	-11.49	0.72
New York	74.36	-10.53	0.21
North Carolina	75.59	-13.04	0.30
North Dakota	81.81	-9.96	0.85
Ohio	79.59	-13.87	0.24
Oklahoma	75.78	-14.89	0.45
Oregon	79.46	-12.51	0.44
Pennsylvania	79.29	-12.78	0.22

Rhode Island	72.94	-11.93	0.94
South Carolina	75.57	-13.11	0.43
South Dakota	81.70	-10.39	0.85
Tennessee	77.03	-13.52	0.36
Texas	72.62	-11.76	0.20
Utah	81.16	-11.26	0.54
Vermont	78.58	-11.05	0.93
Virginia	79.01	-12.04	0.31
Washington	80.05	-11.63	0.33
West Virginia	78.92	-12.12	0.75
Wisconsin	83.17	-13.42	0.27
Wyoming	77.83	-11.64	1.37

Source: 2020 Decennial Census Unedited File and 2015 to 2019 American Community Survey Address File. State self-response rates in this chart may not match other published census numbers given sample restrictions. See note from Table 1 for more details.